

2020 AMC 10A Problem 5 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 5. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 5

Problem:

What is the sum of all real numbers x for which

$$|x^2 - 12x + 34| = 2?$$

Answer Choices:

- A. 12
- B. 15
- C. 18
- D. 21
- E. 25

Solution:

Split the equation into two cases, where the value inside the absolute value is positive and nonpositive.

Case 1:

The equation yields $x^2 - 12x + 34 = 2$, which is equal to $(x - 4)(x - 8) = 0$. Therefore, the two values for the positive case are 4 and 8.

Case 2:

Similarly, taking the nonpositive case for the value inside the absolute value notation yields $x^2 - 12x + 34 = -2$. Factoring and simplifying gives $(x - 6)^2 = 0$, so the only value for this case is 6.

Summing all the values results in

$$4 + 8 + 6 = \boxed{(C) 18}$$

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